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Industrial Pharmacy (Phar 511)
Laboratory Manual



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INDUSTRIAL PHARMACY (PHAR 511)
PRACTICAL SESSIONS

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EXPERIMENT 1

Micromeritics: Determination of Particle Size by Microscopy

Objective:

To demonstrate the methods and techniques used in determination of particle size using microscopy.

Background:

Micromeritics is a science and technology of small particles, such as powders and granules. Knowledge of the behavior of small particles is important for the understanding of the properties of the various pharmaceutical preparations. The study of particle size and size distribution that governs the surface area of particles gives valuable information about the dosage forms. By studying and manipulating those properties we can modify the physical, chemical and pharmacological properties of a drug. For example, increasing the dissolution or release of drug from dosage form can be achieved by increasing the surface area. Successful formulation of suspension, emulsion and tablet both from the view point of physical stability and pharmacological response are possible. Control of the particle size is essential in achieving the necessary flow properties of granules and powders.

Particle size distribution and surface area are the two fundamental properties of any collection of particles. There are, in addition to these, numerous derived properties that are based upon these fundamental properties. Derived properties of particular importance to pharmacy include porosity, flowability, density and bulkiness.

Powder when observed under naked eye seems uniform, but they are heterogeneous. The methods used to determine particle size includes microscope, sieving, sedimentation, coulter counter and light diffraction.

In particle size analysis the microscope is extremely useful. It makes possible not only to measure the size of the particle but also to see and study its shape and other surface features. A good laboratory microscope can be used to see particles having range of

sizes from 0.2 to 100 microns. An ultra microscope can detect particles as small as 0.01 microns and an electron microscope can reveal particles in the angstrom range.

Procedure:

Measuring particle size with an ordinary microscope includes the following steps:

1. Selection of a representative sample.
2. Standardization of the ocular micrometer
3. Preparation of slide containing the sample
4. Adjustment of the ocular system to give resolution
5. Selection of particles to be studied. A statistically valid selection should be made
6. Actual measurement of the particles and group them in various size classes
7. Counting of particles in each classes
8. Analysis of the data

200 to 500 particles should be counted in order to have statistically valid result.

Standardization of Ocular Micrometer

The Ocular micrometer is standardized in terms of stage micrometer for a particular magnification. A stage micrometer is a slide that contains a division of known dimension and used to measure the side of the square on the ocular eyepiece.

Preparation of the Slide:

Approximately 1 mg of a powder is placed on a slide and a drop of redistilled turpentine oil or other non-solvent volatile liquid is added. The powder is then rubbed gently back and forth into the liquid with the use of a smooth glass rod until the turpentine is volatilized, leaving the particles fixed on the slide. This step requires a high degree of precision, as particle size reduction could be possible.

Measurement of the Particles:

The size of about 300 particles should be measured and the results recorded. The graphical representation of microscopic analysis is made in three ways.

The percent frequency is plotted against the logarithm of particle size in microns.
 The cumulative percent frequency undersize and oversize is plotted against the particle size in microns.

The logarithm of particle size is plotted against the cumulative percent undersize on probability scale (log probability plot). From the log probability plot, geometric mean diameter (d_g) and geometric standard deviation (δ_g) will be calculated.

Results:

Size range in μm	Mean size range (d) in μm	Number of particles in each size range (n)	Percent	Cumulative % frequency undersize	Cumulative % frequency oversize

Discussion:

1. Calculate the Length-Number mean diameter and Volume Surface diameter based on the experimental data.
2. Convert the Number Distribution obtained in the experiment into the Weight Distribution and plot the result.
3. From the results of geometric mean diameter (d_g) and geometric standard deviation calculate the length number mean by using relevant Hatch-Choate equation.

EXPERIMENT 2

Micromeritics: Determination of Particle Size by Sieving (Sieve Analysis)

Objective:

To demonstrate the methods and techniques in determination of particle size by sieving

Background:

A sieve analysis is the simplest and still one of the most widely used technique in the determination of particle size and size distribution. Many industries utilize this method as a routine control procedure in the manufacturing of products in the powder form.

Placing a weighed amount of the sample material on the top of the sieve nest carries out the sieve analysis. The sieve nest is arranged by placing the sieve with the largest sieve opening on the top and arranging the remaining sieves in decreasing order of mesh size. The pile of sieves with the sample on the top sieve is shaken in a definite manner either manually or mechanically for a definite time. The material retained on each sieve is collected and weighed.

Procedure:

The sieves are weighed on a balance to the nearest 0.5gram and stacked vertically with the coarsest sieve on the top. Hundred grams of the sample of the material to be analyzed is placed on the top sieve and shaken for 15 minutes. The sieves are separated and weighed individually. The amount of sample retained on each sieve is obtained by subtracting the empty weight of the sieve.

The graphical representation of the sieve analysis is made in three ways:

1. The weight percent frequency is plotted against the logarithm of average particle size in microns.
2. The cumulative percent frequencies undersize and oversize are plotted against the particle size in microns.
3. The log of the sieve apertures plotted against the cumulative percentage undersize and the plot obtained is called log-probability plot.

Results:

Mesh number	Weight of sieve (grams)	Weight of sieve + powder retained after sieving (grams)	Weight of powder on sieve (grams)

Screen opening (μm)	Average weight of particles retained (grams)	Percent (%) of particles retained	Cumulative Percent oversize	Cumulative Percent under size

EXPERIMENT 3

Micromeritics: Determination of Some Derived Properties of Powders and Granules

Background:

The derived properties of powders and granules include the flowability, density and porosity.

1. Flowability

Particles may be free flowing or cohesive. The free-flowing property of particles is desirable in pharmaceutical processes. The flowability of a powder or granulation may be expressed as a rate of flow or angle of repose. The flow property depends on the frictional forces between the particles. When the frictional force is increased, as in the case of rough and irregular particles, the flowability decreased. The frictional force between particles may be expressed in terms of the angle of repose, which is defined as the maximum angle possible between the surface of a pile of powder and the horizontal plane. However, pharmaceutical processes are concerned with the actual movement of particulate systems in mechanical equipment, therefore, the rate of flow is more meaningful than the static property characterized by the angle of repose. The rates of flow can be determined by weighing the material that has flowed in a fixed time interval through a circular orifice and is expressed in gm sec^{-1} .

Procedure

Measurement of Angle of Repose

A funnel with the end of the stem out perpendicular to the axis of symmetry is clamped with its tip at a specified height (H) above a graph paper placed on a float horizontal surface. About 100gms of granules are carefully poured through the funnel and the tip is released till the apex of the conical pile formed just touches the tip of the funnel. The Mean diameter (2R) of the base pile of the granules is determined and the tangent of angle of repose is calculated.

Density

Density is defined as weight per unit volume. The weight of a powder is determined easily and unambiguously. Difficulty arises when one attempts to determine the volume of particles containing microscopic cracks, internal pores and capillary space. Considering the types of powder volume, there are three types of densities: bulk, granule and true density.

A. Bulk density

Bulk density is determined from the bulk volume measured in graduated cylinder. The bulk density of a powder depends primarily on particle size distribution, particle shape and tendency of the particles to adhere together. The particles may pack in such a way as to leave large gaps between their surfaces resulting in a light powder, or powder of low bulk density. On the other hand, the smaller particles sift between the larger ones to form a heavy powder or of high bulk density. Bulk density may be used to check the uniformity of bulk chemicals, and to determine the proper size of containers, mixing apparatus and capsules for a given mass of powder.

Procedure: Measurement of Bulk density

A sample of granules is slowly introduced without pressure into a 100 ml measuring cylinder. The cylinder is tapped on a hard wooden surface from a height of 2 inches, 3 times in two-second intervals. The bulk density is obtained by dividing the weight of the material in grams to the final volume of the samples in the measuring cylinder.

B. Granule density

Granule density is calculated from a volume of the powder obtained by displacement of liquid. The liquid penetrates the intra particle pores, but not internal pores and capillary cracks.

Procedure: Determination of Granule density

A 25ml pycnometer is thoroughly cleaned, dried and weighed. Two or three grams of the powder is placed in pycnometer and benzene of specific gravity 0.879 is employed as the immersion fluid. Weight of the empty pycnometer, pycnometer + benzene and pycnometer + granule + benzene are taken and the granule density is calculated.

Tableting by wet granulation method

Experiment 4 and 5: Tableting

Direct Compression and Wet Granulation

Objective:

The objective of this experiment is to demonstrate the process of direct compression and wet granulation methods.

Background:

The characteristics of a tablet that make it a popular dosage form, such as compactness, rapid production capability, chemical and physical stability, are in general dictated primarily by the qualities of the granulation form. Materials intended for compaction into tablet must be of good flow property (fluidity) and compressibility. An ideal physical shape for good flow property is the spheres. Shapes that approach sphere improve flow property. Spheres offer minimum contact surfaces between particles and the walls of the machine parts thereby reduce the friction force. Granulation is therefore a pharmaceutical process that attempts to improve the flow property and compressibility of powdered materials by forming sphere like aggregates of powders called granules.

There are three basic methods of preparation of compressed tablets:

1. Wet Granulation Method
2. Dry Granulation Methods
3. Direct Compression Methods

Direct compression is a method of tablet making in which

- Directly compressible crystalline drugs are directly compressed without prior granulation step or
- Powdered drugs (low dose) are combined with a pregranulated or coarse particulate diluent and the mixture is directly compressed.

The great advantages of direct compression are the simplicity of the process, avoidance of a granulation step, avoidance of moisture and drying step, minimal material handling, rapidity of the total process and optimum possible bioavailability of drug from the resulting tablets.

Direct compression has distinct limitation; only few crystalline drugs and low dose drugs (with direct compression diluents) lend themselves to direct compression. Examples of drugs that lend themselves to direct compression are sodium chloride, aspirin, potassium chloride, potassium bromide and ammonium sulfate.

Wet granulation method involves dampening or wet massing of the powder mixture with a liquid-adhesive (granulating agent) followed by wet screening in a classical method, the individual steps in the wet granulation process of tablet preparation include, weighing, milling, sieving, dry powder mixing, wet massing, wet screening, drying, sizing, lubrication and compression. In such pharmaceutical process, it is inevitable to have a loss, which makes it difficult to calculate the final compression weight.

The compression weight is the weight of granules that must be fed into the die-cavity to give tablets containing the nominal amount of the active drug. The most direct method is to assay the granules and from the result, calculate the compression weight. However, there is an alternative method of calculating the compression weight from the data collected during the manufacture of the granules as illustrated by the following example.

Example:

Data obtained during granulation for the manufacture 100 tablets each containing 500mg of drug.

Medicament	50 g
Adhesive granulating agent	10 g
Theoretical weight of the moist mass	60 g
Actual weight of moist granules	54 g
Weight of dry granules	50 g
Weight of granules after screening including 10% fines	46 g

Formula of granules for compression

Screened granules + 10% fines

Disintegrant (5%)

Lubricant (1%)

Total:

46 g

2.30 g

0.46 g

48.76 g

The screened granules will be mixed with disintegrant and/lubricant before compressed into a tablet.

Calculation of the Compression weight:

1. Number of tablets that could be prepared from 50g of medicament

$$50\text{g}/0.5\text{g} = 100 \text{ tablets}$$

2. Weight of Moist mass per tablet

$$60\text{g}/100 = 0.6 \text{ g / tablet}$$

3. Number of tablets that may be prepared from the actual weight of moist granules

$$54\text{g}/0.6\text{g} = 90 \text{ tablets}$$

4. Weight of the dry mass per tablet

$$50\text{g}/90 = 0.55 \text{ g / tablet}$$

5. Number of tablets that may be prepared from the screened granules

$$46\text{g}/0.55 = 83 \text{ tablets}$$

6. The final compression weight after the addition of disintegrant and lubricant.

$$48.76\text{g}/83 = 0.587 \text{ g}$$

Practical Experiment 4

1. Prepare 200 tablets each containing 300mg of microcrystalline cellulose (Avicel) by direct compressive method
2. Compress the tablets at two different compression forces (low and high, e.g. at 2kN and 6kN), and see the effect of compression force on hardness and friability.

Practical Experiment 5

3. Prepare and supply 400 chloroquine phosphate tablets each containing 500mg of the drug using the wet granulation method.

Formula for Chloroquine phosphate tablets

Chloroquine Phosphate
Starch Mucilage
Starch
Magnesium Stearate
Talc, Purified

$\frac{500 \times 400}{50 + 200}$
100g
500 mg
q.s. - 2 way
10% - 5 g - 100 mg
0.25%
0.75%

Procedure:

Prepare 10% starch mucilage by adding 10g of starch in 20ml of water stir until the starch is dispersed in the water. Then adjust the volume to 100ml with hot water and heat until translucent mucilage forms.

Mix the drug with the starch thoroughly in order of increasing bulk. Moisten the powder blend with the 10% starch mucilage until a coherent mass forms. The mass has to be neither too dry nor too wet. The wet mass is then passed through a wet granulator and dried in hot air oven at 50°C.

At each step note the following data to calculate the compression weight of the tablet.

Theoretical weight of the moist mass:

Actual weight of moist granules:

Weight of granules after drying:

Weight of granules after screening (including 10% fines):

As: jnm nt

1. Discuss the reasons behind each step (unit process) in wet granulation process.
2. What are the advantages and disadvantage of wet granulation in tablet production?
3. Explain the effect of excess granulating agent on the resulting granule and tablet properties.

Experiment 6

Compression and Evaluation of Physical Properties of Tablets

Objectives:

The objectives of this experiment are to prepare tablets and to evaluate some of the physical properties of the resulting tablets.

Procedure:

Preparation of granules for compression (continued from the previous experiment)

Screen the dried granules and remove the fine powders keeping only 10% of it. Weigh the required amounts of the magnesium stearate and talc and mix them with the screened granules in a cube-mixer for 5 minutes. Calculate the compression weight of the tablets from the previous data.

Compression of granules

Assemble the appropriate size of punches, die and other parts of the instrumented eccentric EK-0 tablet machine correctly. Put the lubricated granules in the hopper and adjust the tablet weight by changing the die fill volume. Once the fill volume is adjusted, start the machine, first manually and then automatically (using the motor); adjust the compression force to 15KN.

Evaluation of the physical properties of the tablets

Evaluate the compressed chloroquine phosphate tablets for their *weight variation, hardness, friability and disintegration times* using standard methods.

Weight Variation Test

Weigh individually 20 whole tablets and calculate the average weight. Record the weight of the individual tablets. The weights of not more than 2 of the tablets differ from the average weight by more than the percentage listed and no tablet differs by more than double that percentage (look up the weight variation tolerance in the BP and USP).

Friability Test

20 tablets are weighed accurately and placed in ERWEKA friability tester. The motor is run at a speed of 25 rpm for a period of 4 minutes. After this, the dust is removed and the tablets are reweighed. Compressed tablets that lose less than 0.5 – 1.0% in weight are considered acceptable.

Hardness/Crushing Strength Test

The crushing strengths of 10 tablets are measured using Schleuniger hardness tester. In this, each tablet is placed between two anvils and the force that causes each tablet to break is recorded as the crushing strength. The average crushing strength is taken and the standard deviation will be calculated.

Disintegration Test of Uncoated Tablets

Place 1 tablet in each of the six tubes of the basket-rack assembly, add a disk to each tube and operate the apparatus, using water maintained at $37 \pm 2^\circ\text{C}$ as the immersion fluid. At the end of the time limit specified in the monograph, lift the basket from the fluid and observe the tablets: all of the tablets have disintegrated completely. If 1 or 2 tablets fail to disintegrate completely, repeat the test on 12 additional tablets. Not less than 16 of the total 18 tablets tested disintegrate completely.

Assignment

1. Sketch the EK-O eccentric tablet machine.
2. Briefly discuss the advantages of instrumentation.

Experiment 7

② Determination of Dissolution Rates of Tablets

Objectives:

The objective of this experiment is to evaluate the dissolution rate of the chloroquine phosphate tablets.

Procedure:

Preparation of Calibration Curve

Prepare a stock solution of the pure (chloroquine phosphate) drug using 0.1N HCl (pH 1). From the stock solution, prepare six different concentrations by serial dilution. Measure the absorbances of the solutions using a UV-Spectrophotometer at the λ_{max} of the drug (344 nm). It is preferable if the absorbance reading of the six solutions is within the range of 0.200 and 0.800. Plot the Beer's Lambert calibration curve and derive the linear regression equation.

Dissolution Test

Place 1000ml of 0.1N HCl (pH 1) in each vessel of the dissolution apparatus; assemble it and equilibrate the dissolution medium to $37 \pm 1^\circ\text{C}$. Place 1 tablet in each basket and immediately operate the apparatus at a rotational speed of 100rpm. Withdraw 10ml samples at predetermined time intervals (5min, 10min, 15min, 20min, 30min, 45min, and 60min). Sampling is done midway between the surface of the dissolution medium and the top of the rotating basket not less than 1cm from the vessel wall. Filter each sample and, after the necessary dilution, read the absorbance using the UV-Spectrophotometer at the λ_{max} . The concentration of the drug in each sample is determined from the Beer's Lambert calibration curve constructed previously for the pure drug. Calculate the cumulative percent release of chloroquine phosphate from the tablets with time and plot the dissolution curve.

NB:

- Replace sample volumes after each sampling with equal volumes of dissolution medium
- Do the necessary corrections for the dilutions made when calculating for the amount of drug released at each sampling time.

Assignment

Comment whether the dissolution of the tablets comply with pharmacopoeial specifications (Refer to the specifications in the BP and USP).